

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 3: Short Answer Test**

**COURSE NAME:** Cloud Computing and  
Big Data Analytics

**COURSE CODE:** CSA1522

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**COURSE OUTCOME COVERED**

**CO1:** Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)  
Question Count: 5

**Total Marks: 50**

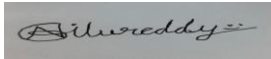
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**Code of Conduct**

I G. VENKATA NIRANJAN REDDY certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



**Assessment Tasks**

**Short Answer Test**

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

**Short Answer Test Rubric**

| Criteria             | Weightage | Level 4 - Exemplary                                           | Level 3 - Proficient            | Level 2 - Developing                         | Level 1 - Beginning                 |
|----------------------|-----------|---------------------------------------------------------------|---------------------------------|----------------------------------------------|-------------------------------------|
| Conceptual Accuracy  | 30%       | Accurate definitions and precise explanation of all key terms | Mostly accurate with minor gaps | Partial understanding with noticeable errors | Incorrect or irrelevant explanation |
| Coverage of Concepts | 20%       | Covers all expected sub-points completely                     | Covers most expected points     | Covers only basic points                     | Very limited coverage               |

|                              |     |                                                              |                                               |                                           |                                 |
|------------------------------|-----|--------------------------------------------------------------|-----------------------------------------------|-------------------------------------------|---------------------------------|
| Use of Examples              | 15% | Uses highly relevant cloud examples to strengthen the answer | Uses relevant examples with minor mismatch    | Uses vague or partially relevant examples | No useful examples              |
| Comparison / Differentiation | 20% | Clearly distinguishes concepts with strong logic             | Reasonable differentiation with minor overlap | Comparison is weak or incomplete          | Fails to differentiate concepts |
| Clarity of Expression        | 15% | Clear, organized, and concise answer writing                 | Generally clear with few issues               | Understandable but poorly organized       | Difficult to follow             |

## **ANSWER 1:**

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and processing power over the internet ("the cloud"). It allows users to access these resources on demand without owning or maintaining physical infrastructure.

The four characteristics that make it different from traditional on-premise computing are:

1. **On-Demand Self-Service**

Users can access computing resources such as virtual machines or storage whenever they need them without requiring assistance from the service provider.

Difference from on-premise: In traditional computing, IT staff must install, configure, and allocate hardware before it can be used.

2. **Broad Network Access**

Cloud services are available over the internet and can be accessed from various devices, including laptops, smartphones, and tablets.

Difference from on-premise: Traditional systems are often limited to a company's local network or require special remote access.

3. **Rapid Elasticity (Scalability)**

Cloud resources can be increased or decreased quickly based on demand. For example, additional storage or computing power can be added instantly during peak usage.

Difference from on-premise: Scaling traditional infrastructure requires purchasing, installing, and configuring additional hardware, which is time-consuming and expensive.

4. **Measured Service (Pay-as-You-Go)**

Cloud providers monitor resource usage, and customers pay only for the resources they consume.

Difference from on-premise: Traditional computing requires large upfront investments in hardware and software regardless of actual usage, along with ongoing maintenance costs.

## **Summary:**

Cloud computing provides flexible, internet-based access to computing resources with features such as on-demand self-service, broad network access, rapid scalability, and pay-as-you-go pricing. These characteristics make it more cost-effective, scalable, and convenient than traditional on-premise computing.

## **ANSWER 2:**

### **Evolution of Cloud Computing**

Cloud computing did not develop suddenly. It evolved through improvements in computer hardware, networking, virtualization, and internet-based software.

#### **1. Hardware Evolution**

- Early computers were large and expensive mainframes.
- Later, smaller and more powerful servers and personal computers were developed.
- This increased computing power and reduced hardware costs.

#### **2. Networking Evolution**

- Computers began connecting through local and wide-area networks.
- Resource sharing and communication between computers became possible.

#### **3. Internet Evolution**

- The growth of the internet allowed users to access information and services from anywhere.
- High-speed internet made remote computing practical.

#### **4. Virtualization**

- Virtualization allowed one physical server to run multiple virtual machines.
- This improved resource utilization and made computing resources easier to manage and scale.

#### **5. Internet-Based Software**

- Software gradually moved from locally installed applications to web-based applications.
- Users could access software through web browsers without installing it on their own computers.

## **6. Web Services and Service-Oriented Architecture**

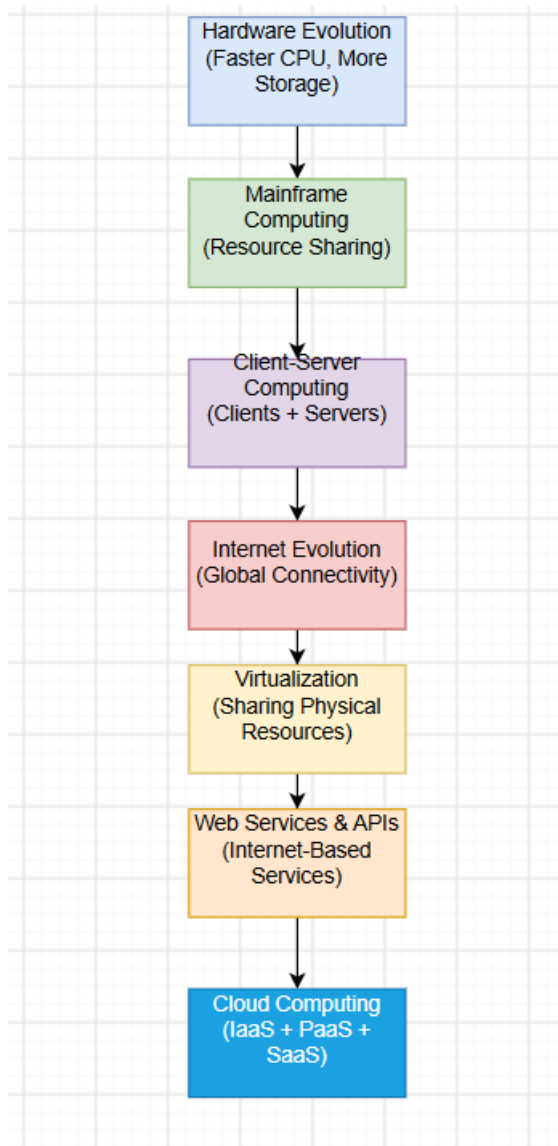
- Applications were divided into reusable services that could communicate over the internet.
- This made it easier to build flexible and distributed applications.

## **7. Modern Cloud Platforms**

- The combination of powerful hardware, virtualization, high-speed internet, automation, and web services led to modern cloud platforms such as AWS, Microsoft Azure, and Google Cloud.
- These platforms provide on-demand computing, storage, databases, networking, and software.

### **Evolution Flow**

**Mainframes → Personal Computers → Computer Networks → Internet →  
Virtualization → Web Applications → Web Services → Cloud Computing**



### ANSWER 3:

#### Role of Server Virtualization in Cloud Computing

Server virtualization is the process of dividing one physical server into multiple virtual machines (VMs) using software called a hypervisor. Each VM can run its own operating system and applications independently.

- ☐ Improves utilization of physical servers.
- ☐ Reduces hardware and maintenance costs.
- ☐ Allows multiple users to share the same physical infrastructure securely.

**Virtualization** :A technology that creates virtual versions of physical resources.

Uses hypervisors to create virtual machines.

Focuses on efficient resource utilization.

## **Cloud Computing**

A service delivery model that provides computing resources over the internet.

Uses virtualization, networking, storage, and automation to deliver services.

**Virtualization is an enabling technology**, while **cloud computing is a service model** that uses virtualization and other technologies to provide resources on demand.

## **ANSWER 4:**

### **SOAP-Based Web Services**

**Communication Style:** SOAP is a protocol-based service that uses XML messages for communication.

**Data Exchange:** It mainly exchanges data in XML format.

**Cloud Suitability:** SOAP is more complex and heavyweight, so it is less suitable for lightweight cloud applications. However, it is useful for enterprise applications requiring high security and strict standards.

### **REST-Based Web Services**

**Communication Style:** REST uses standard HTTP methods such as GET, POST, PUT, and DELETE.

**Data Exchange:** REST commonly uses JSON, but it can also use XML and other formats.

**Cloud Suitability:** REST is lightweight, fast, flexible, and highly scalable, making it well suited for modern cloud, web, and mobile applications.

## **Conclusion**

**SOAP** is suitable for complex enterprise applications, while **REST** is preferred for modern cloud applications because of its simplicity, flexibility, and scalability.

## **ANSWER 5:**

### **1. IaaS – Infrastructure as a Service**

IaaS provides **virtualized computing resources** such as servers, storage, and networking over the internet.

**Example:** Amazon EC2.

### **2. PaaS – Platform as a Service**

PaaS provides a **platform and development tools** for creating and deploying applications without managing the underlying hardware.

**Example:** Google App Engine.

### **3. SaaS – Software as a Service**

SaaS provides **ready-to-use software applications** over the internet.

**Example:** Gmail.

### **4. XaaS – Anything as a Service**

XaaS provides **various IT services on demand** through the internet, such as storage, security, or databases.

**Example:** Dropbox as **Storage as a Service**.